

C-DOT · 5G DIGITAL TWIN

From synthetic traffic to a defensible control decision

Simulator → Forecaster → Optimizer → MPC / pre-drain → safety gates

588 controller pairs · 125 distribution-blind survival trials

Final evidence story v1 · Static production · candidates shadow/replay only

The digital twin succeeded; controller promotion did not

A useful scientific outcome: engineering readiness and algorithm readiness are different claims

TWIN

- ✓ Causal cohort simulation
- ✓ Bounded streaming memory
- ✓ 12 nodes · 384 shards
- ✓ Frozen, hashed evidence

LEARNING + SEARCH

- ✓ Forecast gains measured
- ✓ Lifecycle survival calibrated
- ✓ Oracle headroom mapped
- ✓ Mechanisms fully exercised

CONTROL

- ✗ MPC benefit unstable
- ✗ Pre-drain failed mixed stress
- ✗ Tail and overflow gates missed
- ✗ Static remains production

DECISION · STATIC IN PRODUCTION · MPC / PRE-DRAIN IN SHADOW OR REPLAY

Each stage answered a different question

The final decision came from progressively harder evidence—not a single benchmark



Early positive signals became hypotheses; larger frozen campaigns were allowed to reject them.

The 5G twin is a closed evidence loop

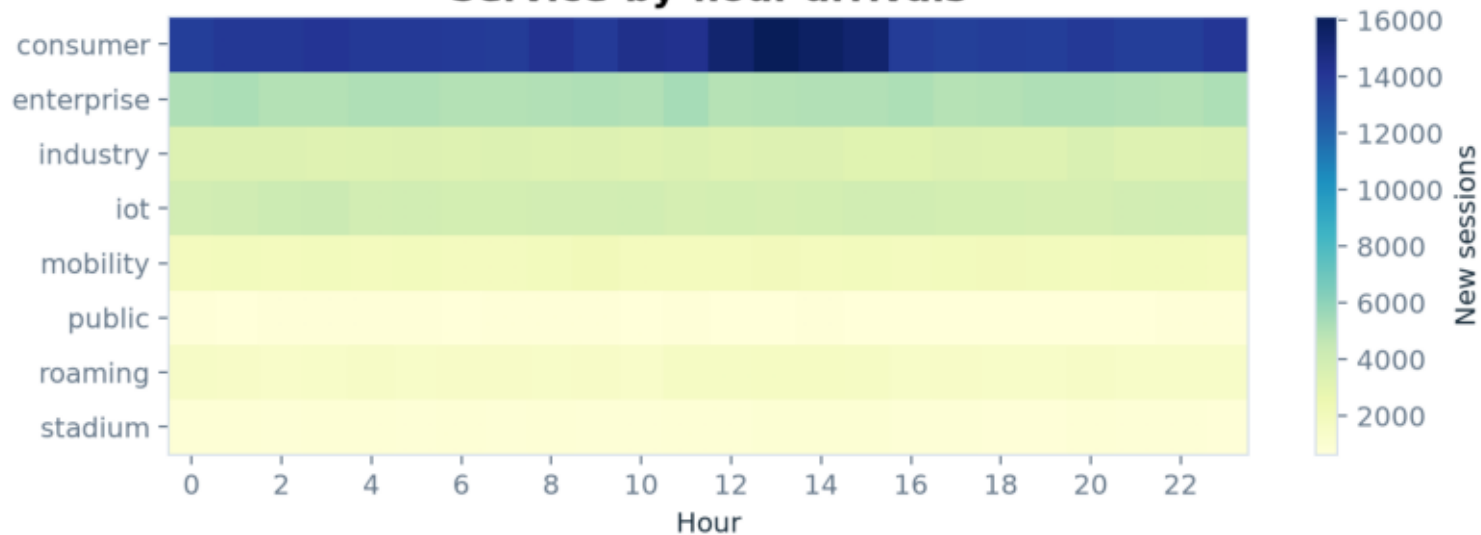
Every proposed action is evaluated against the same causal cohort state before it can affect future sessions



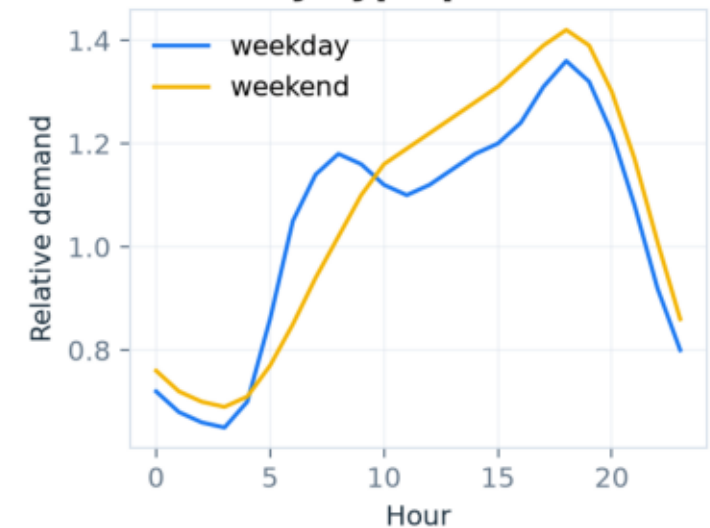
ACTUATION BOUNDARY · EXISTING SESSIONS STAY PUT · ONLY FUTURE SESSIONS CAN MOVE

Population and traffic fingerprint

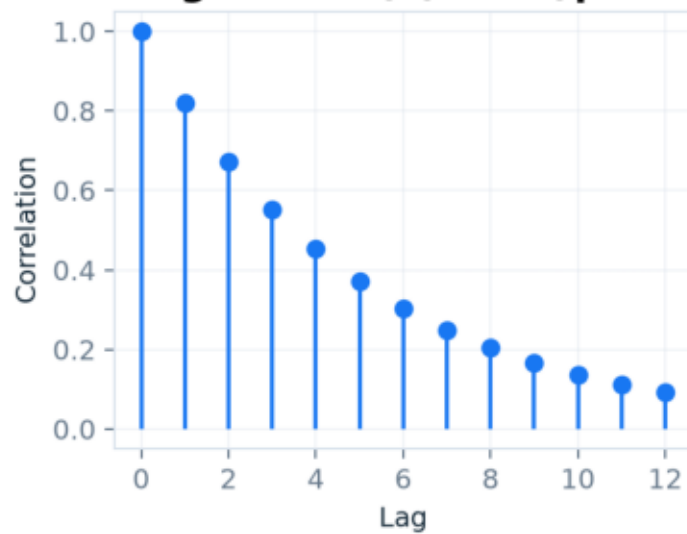
Service-by-hour arrivals



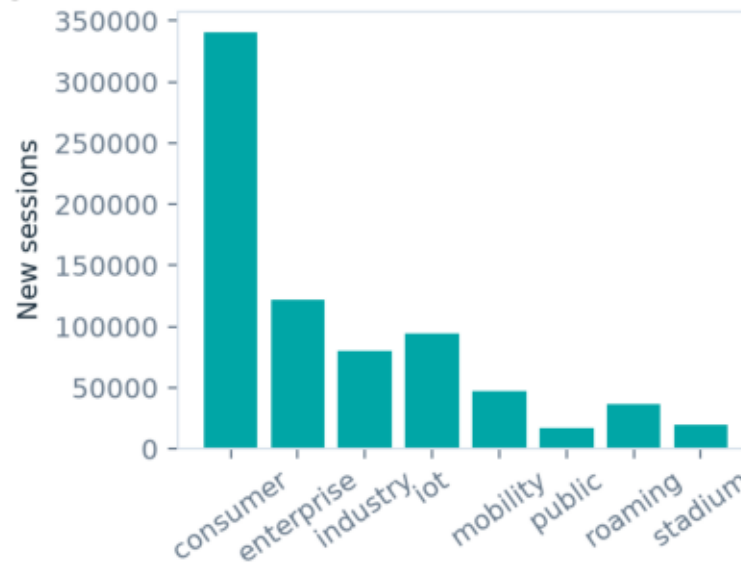
Day-type profiles



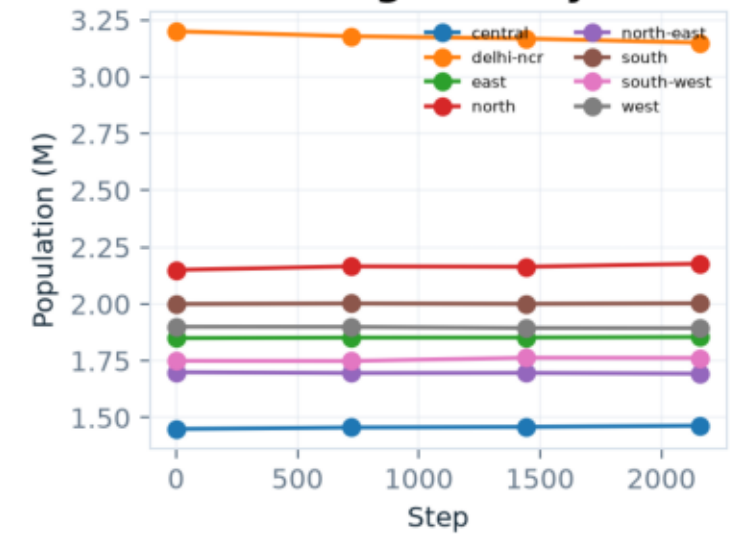
Configured AR(1) ACF ($\phi=0.82$)



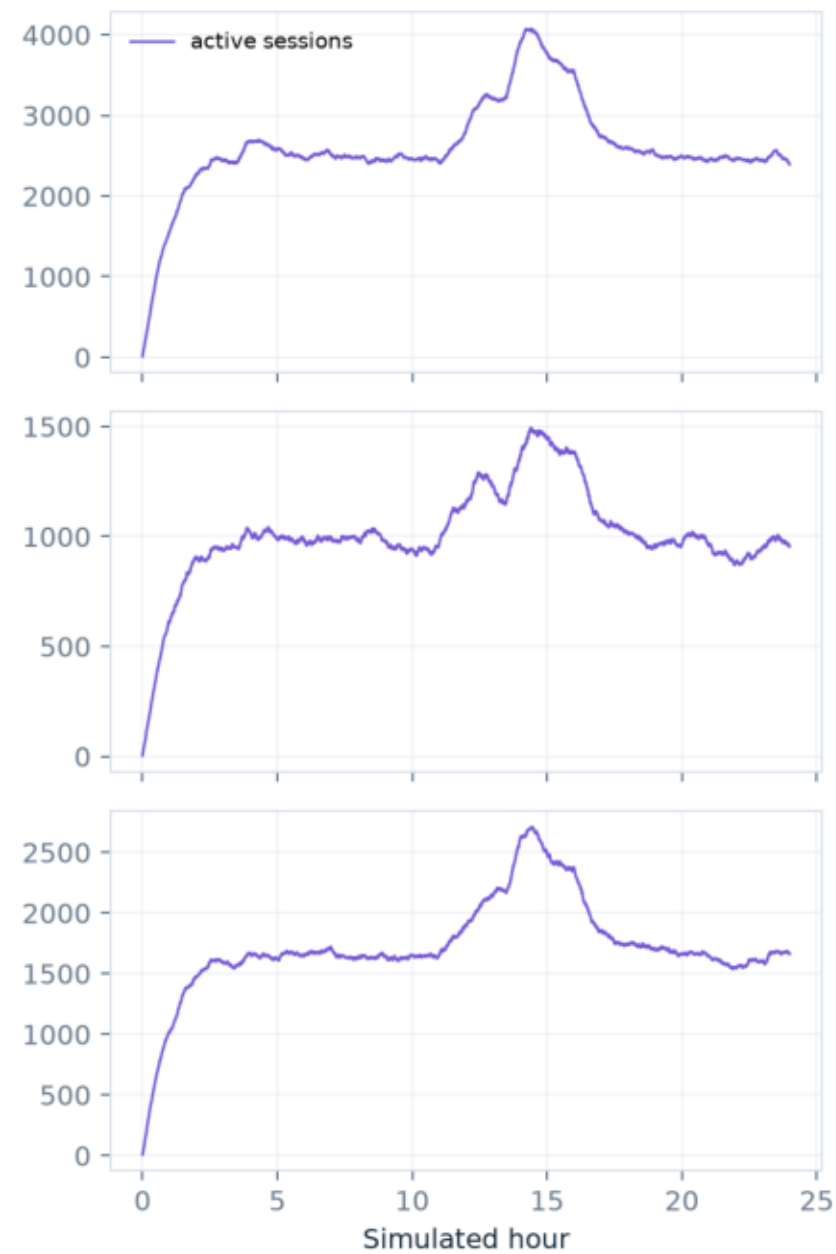
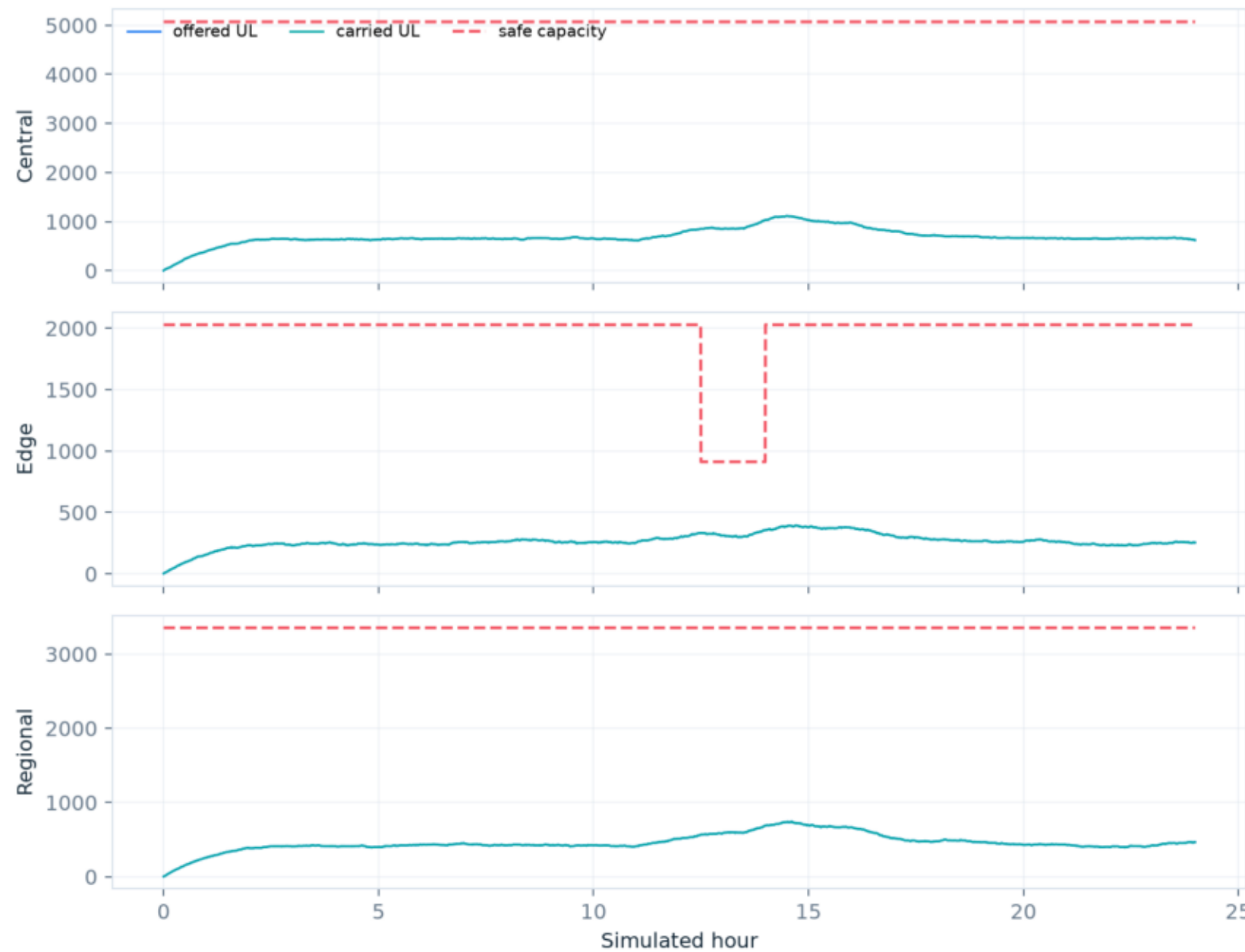
Class mix



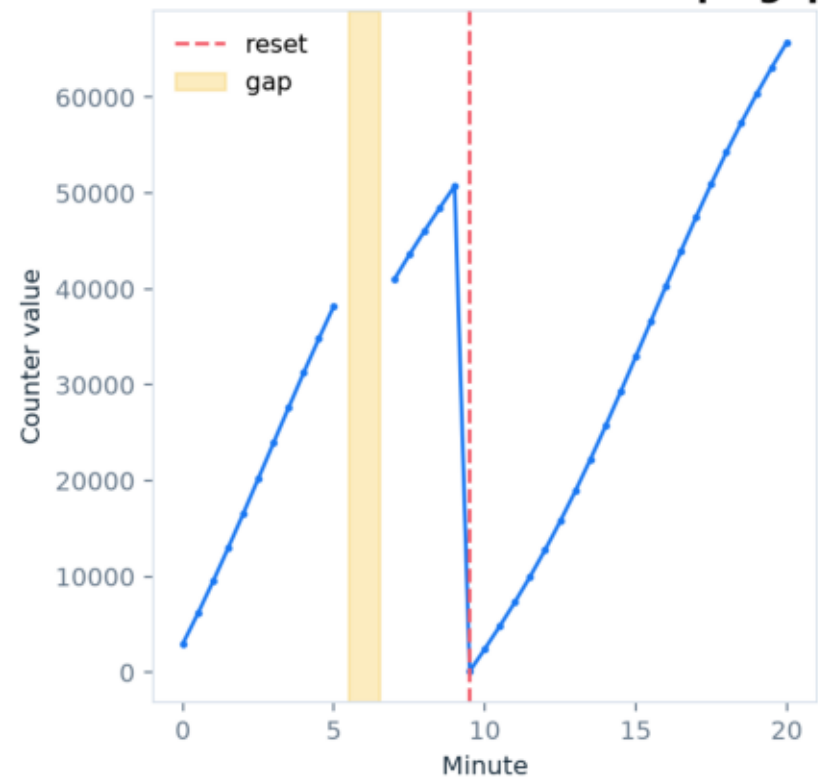
Conserving mobility flows



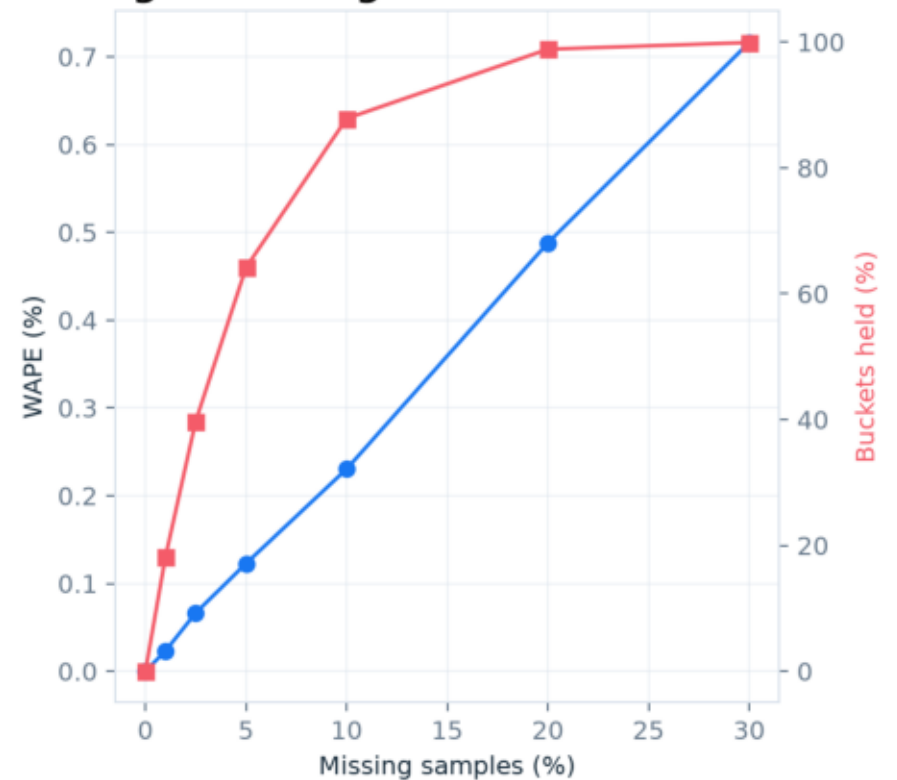
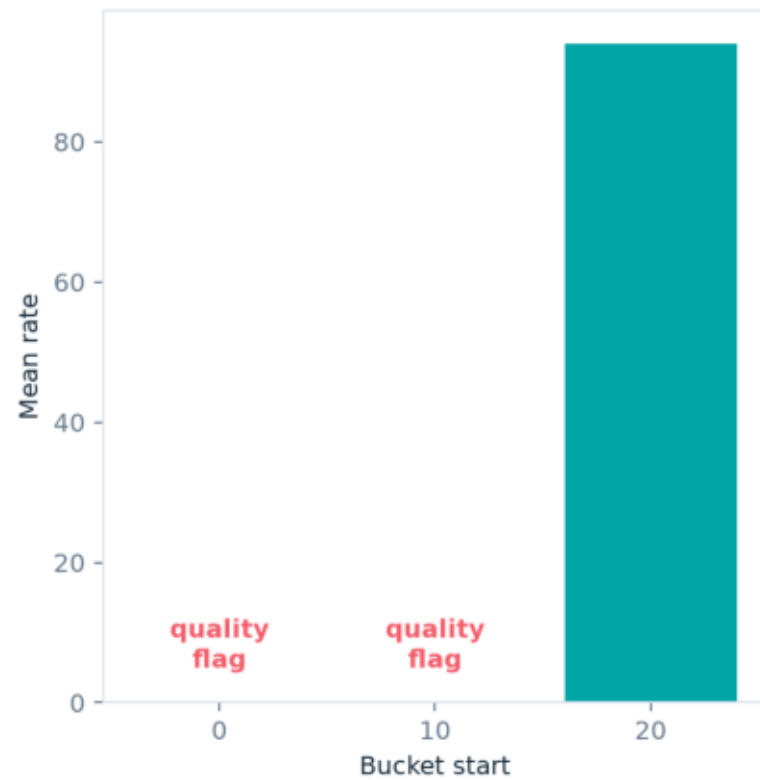
Representative edge / regional / central UPF day



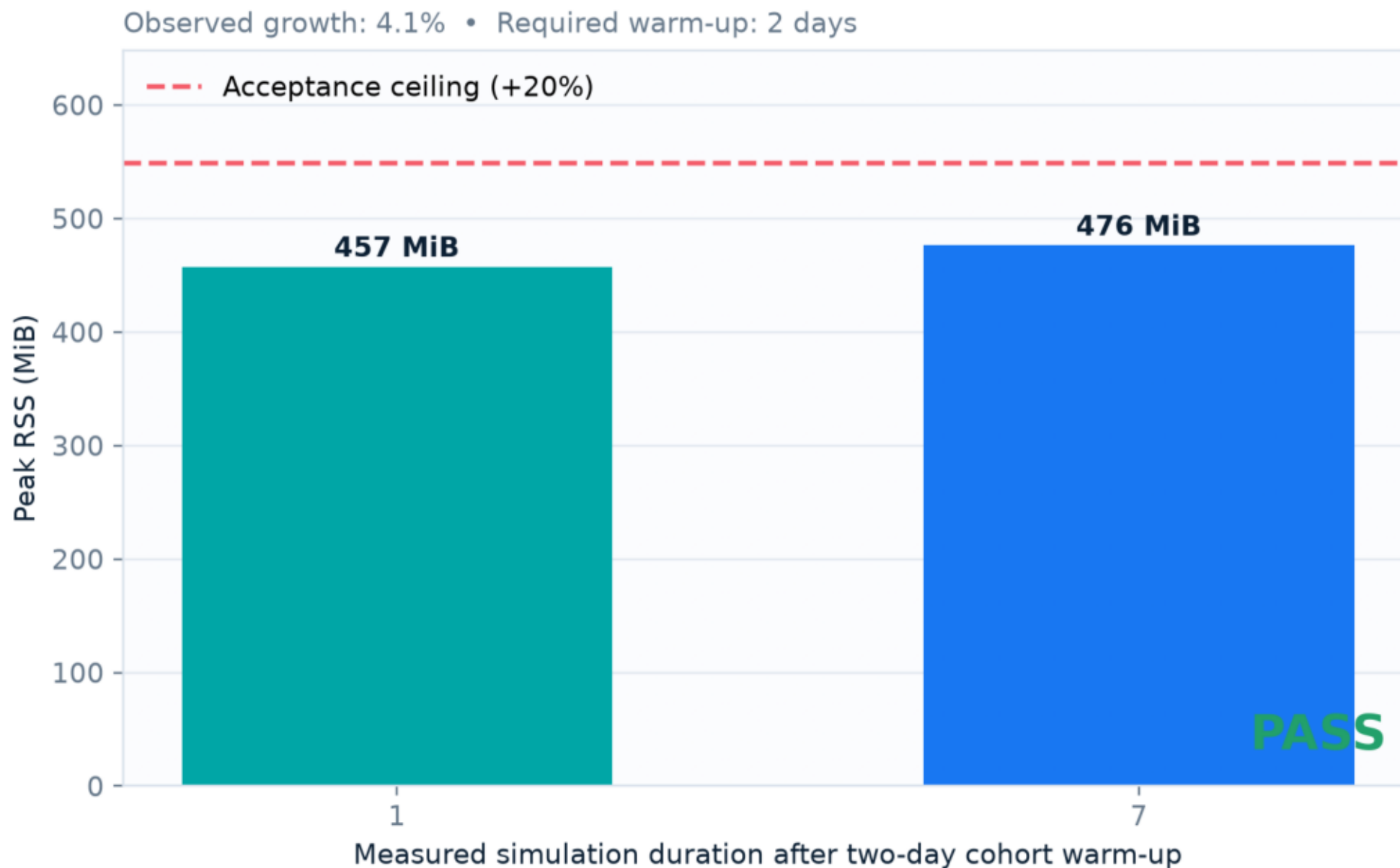
Raw counter: reset and scrape gap



Reconstructed 10-minute buckets Missingness changes estimation and control



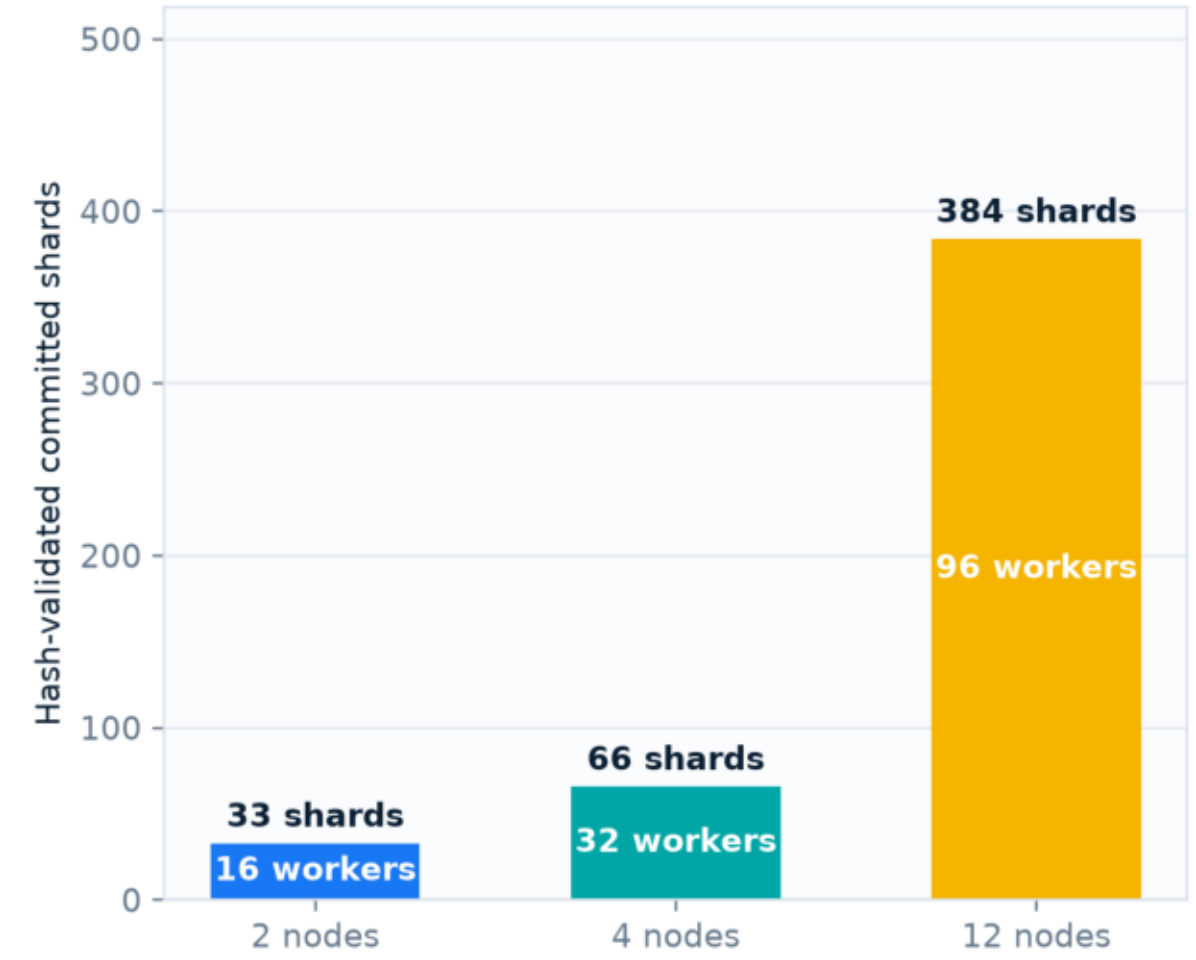
Streaming memory remains flat as duration grows 7×



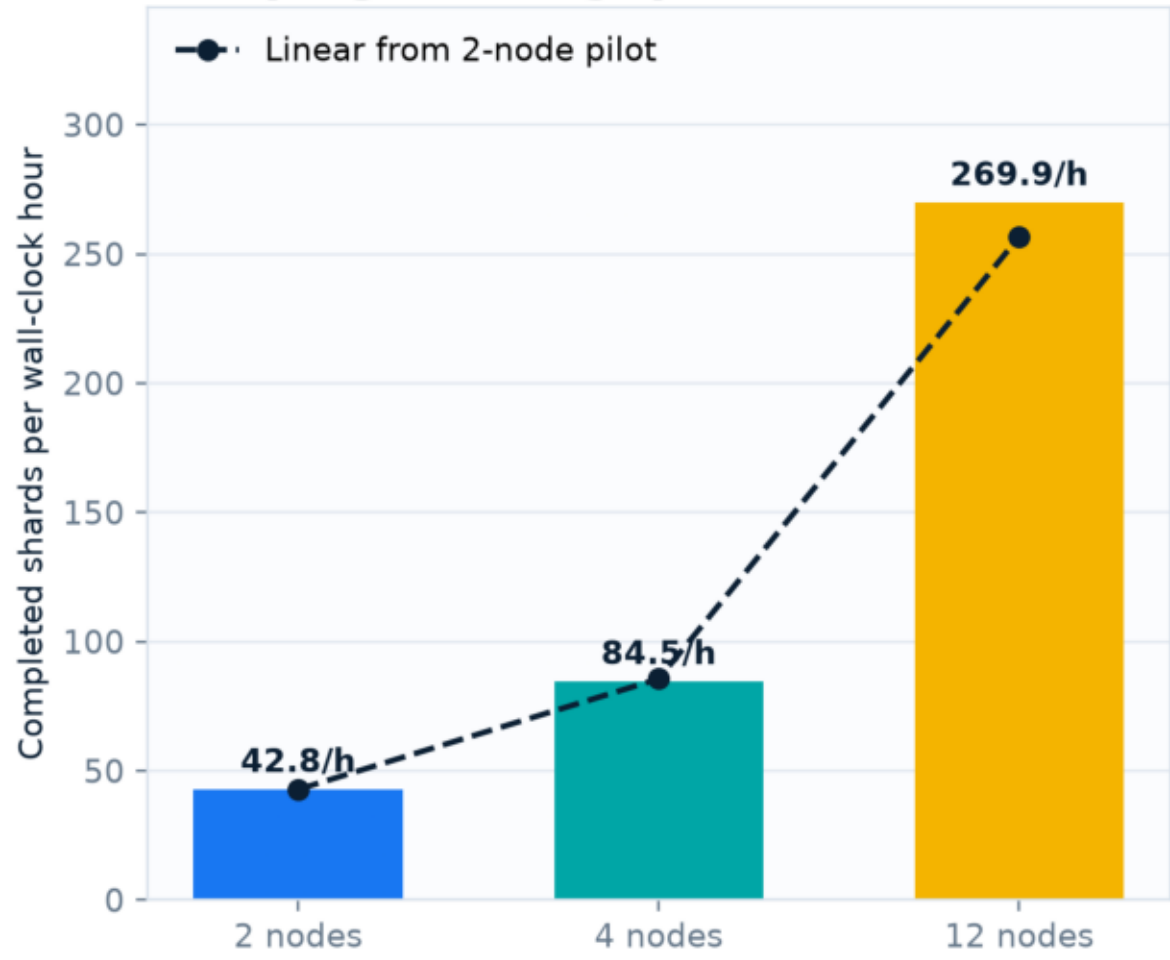
Gated production scales from 2 to 12 nodes

Frozen inputs • balanced mix • zero failures • 12-node run: 4 waves/node; pilots: 2

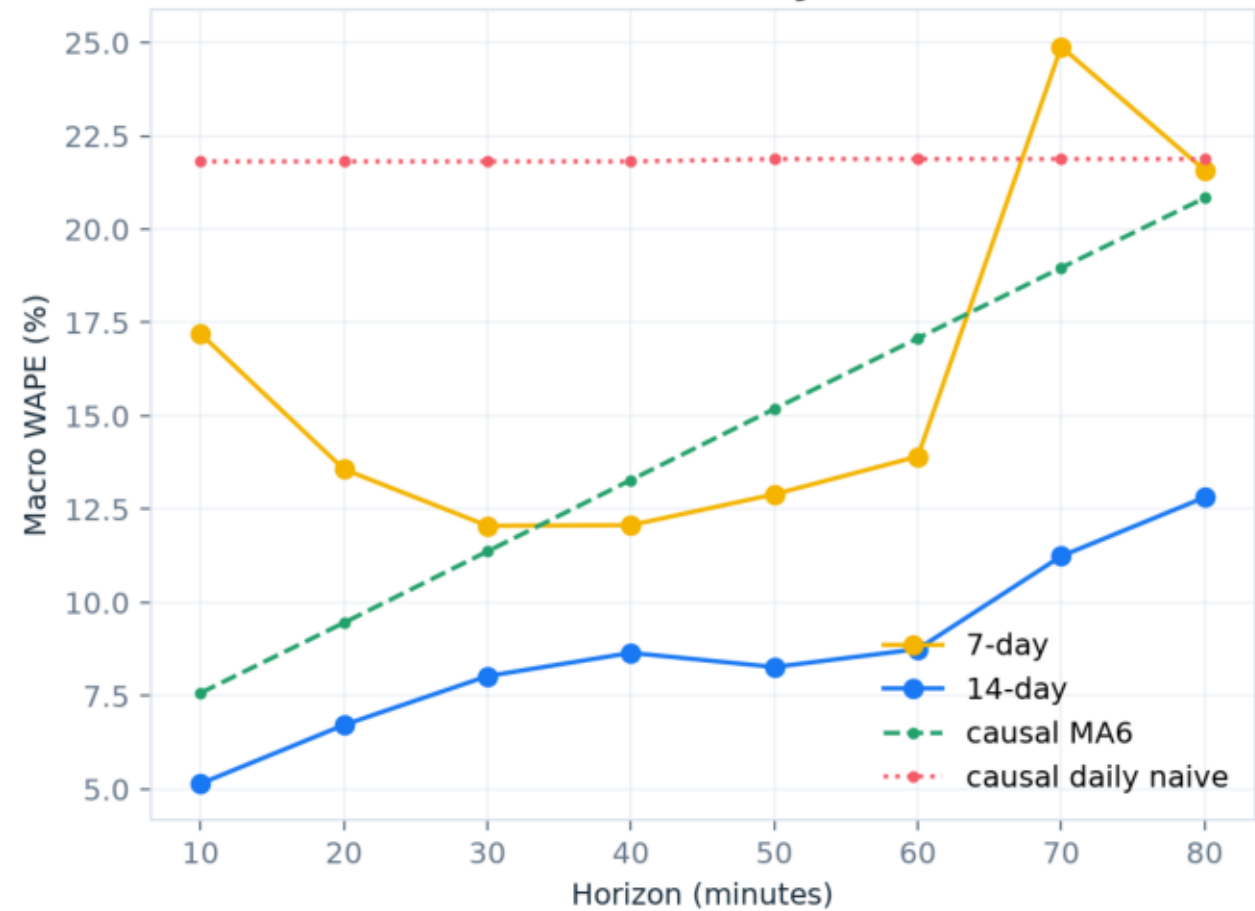
Committed evidence



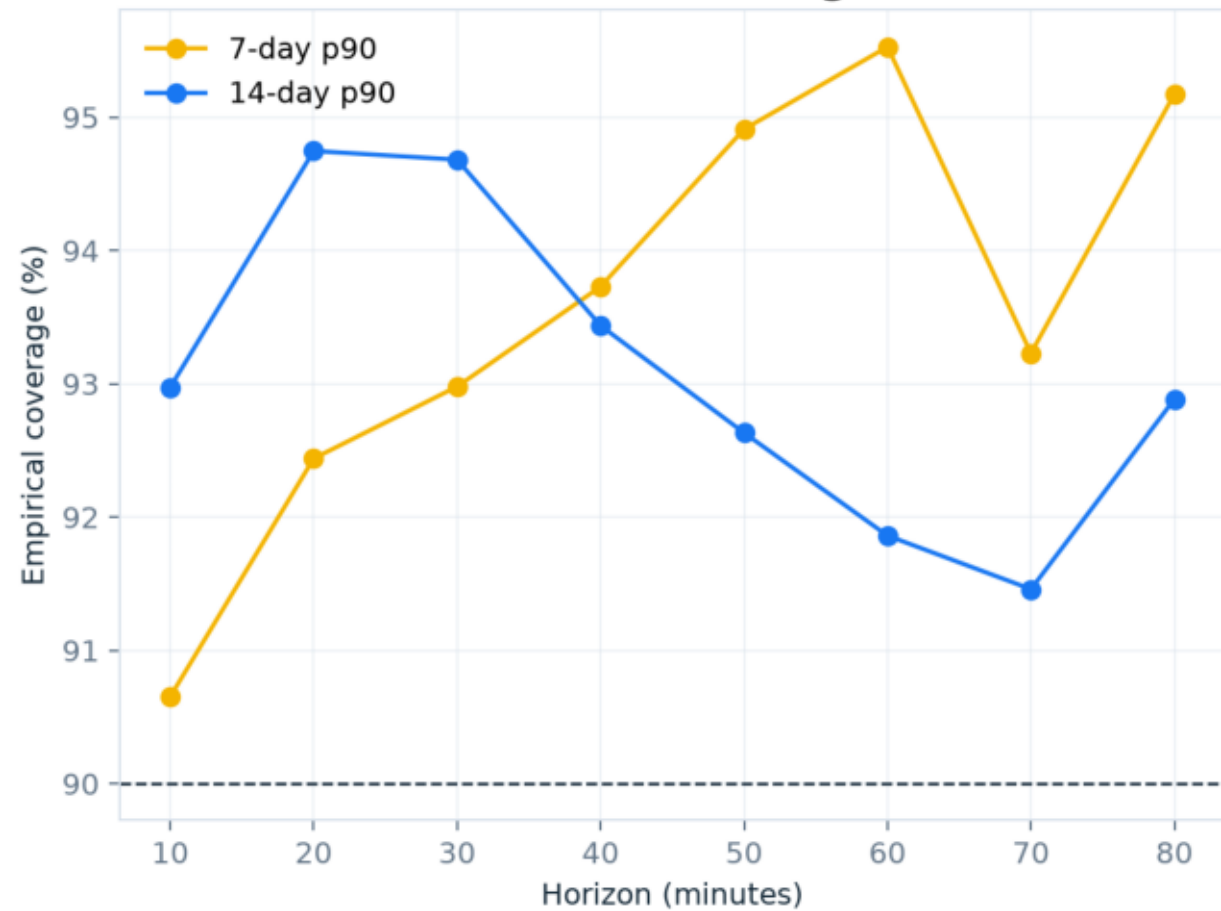
Campaign throughput



Forecast WAPE by horizon

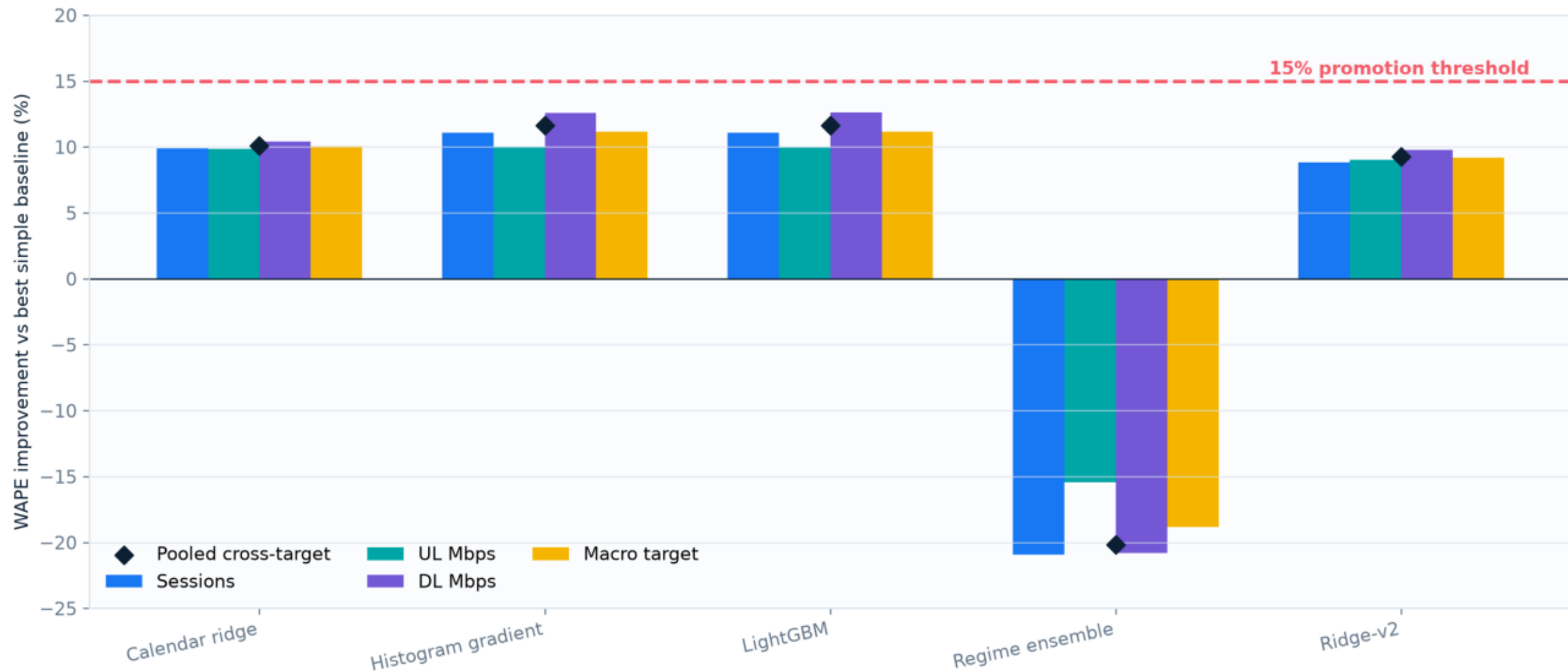


Interval coverage



Forecast gains are real—but below the promotion bar

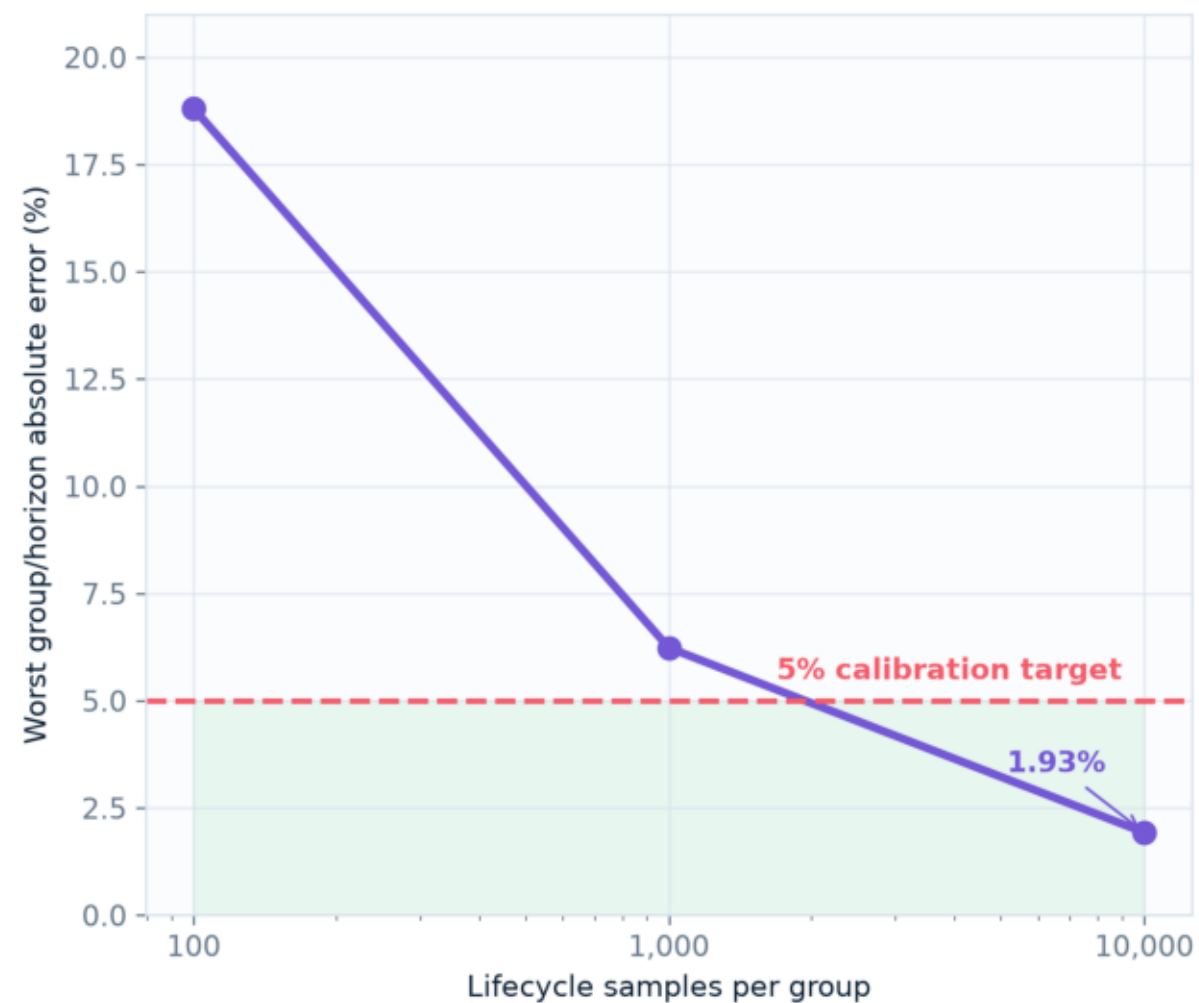
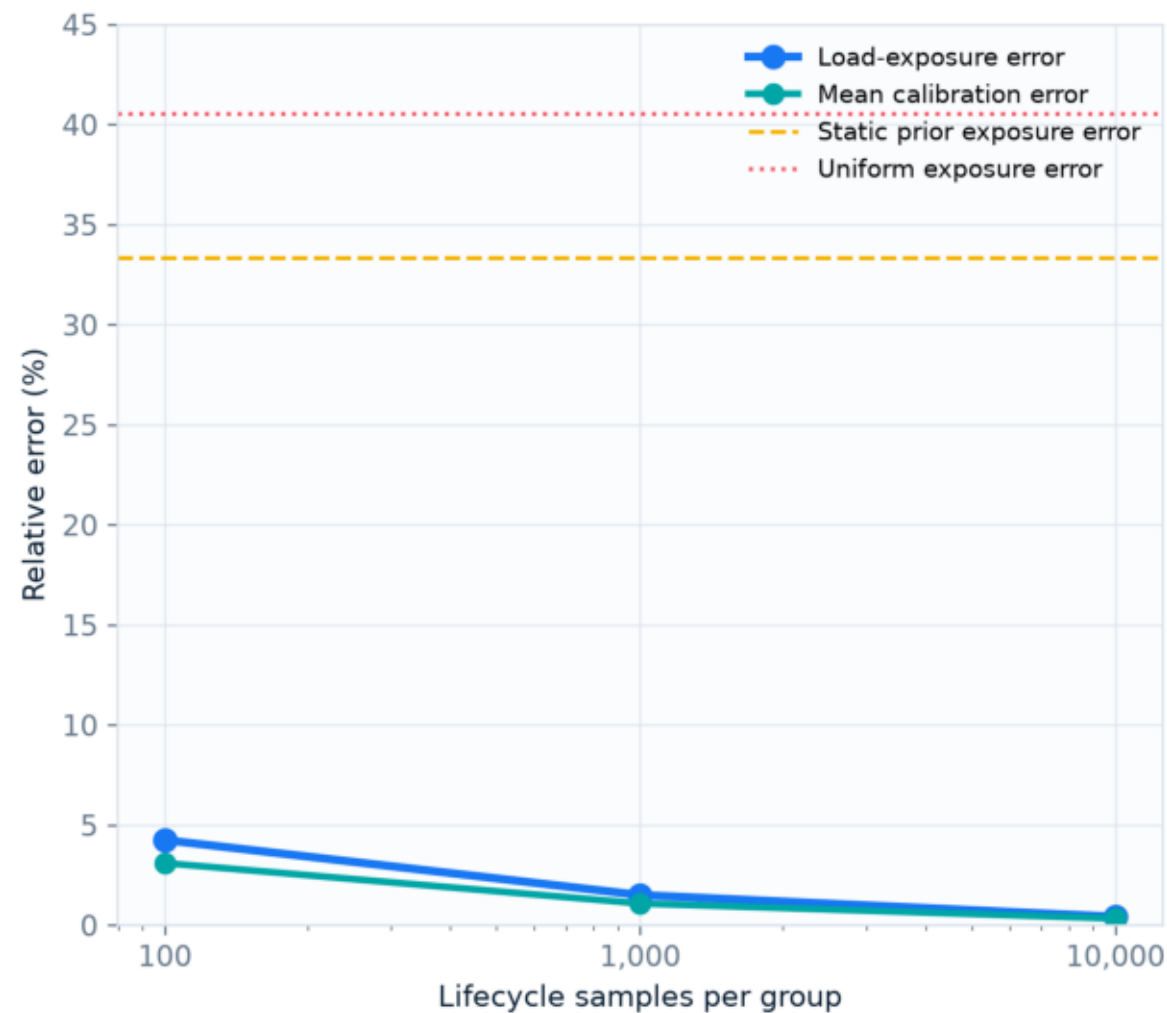
Target-separated WAPE prevents unlike units from being hidden inside the pooled headline



Seed 46003 generated + sealed · never evaluated or selected

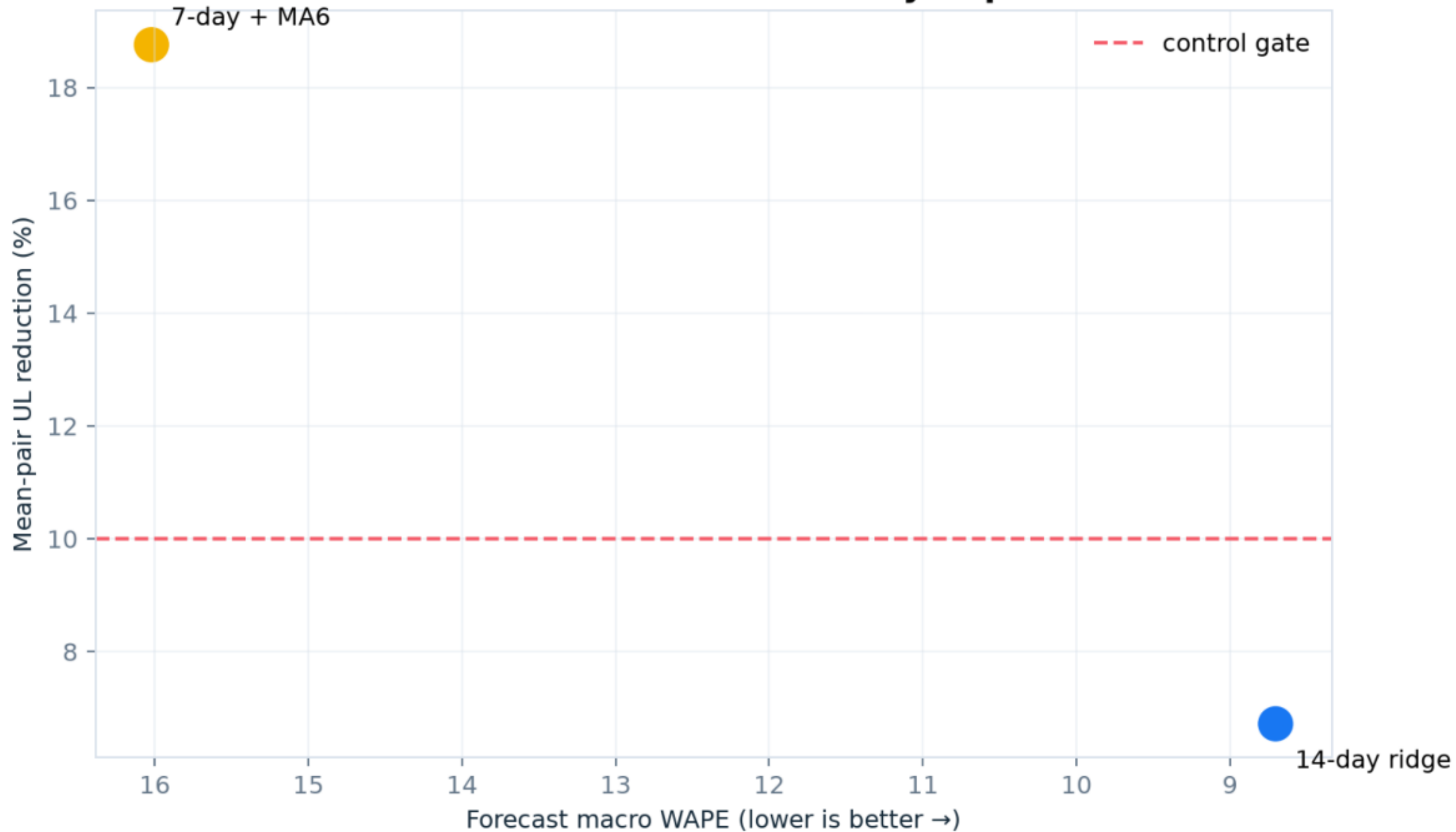
Kaplan-Meier mechanics converge on synthetic lifecycle telemetry

Synthetic censored records test estimation mechanics—not real-world distribution independence

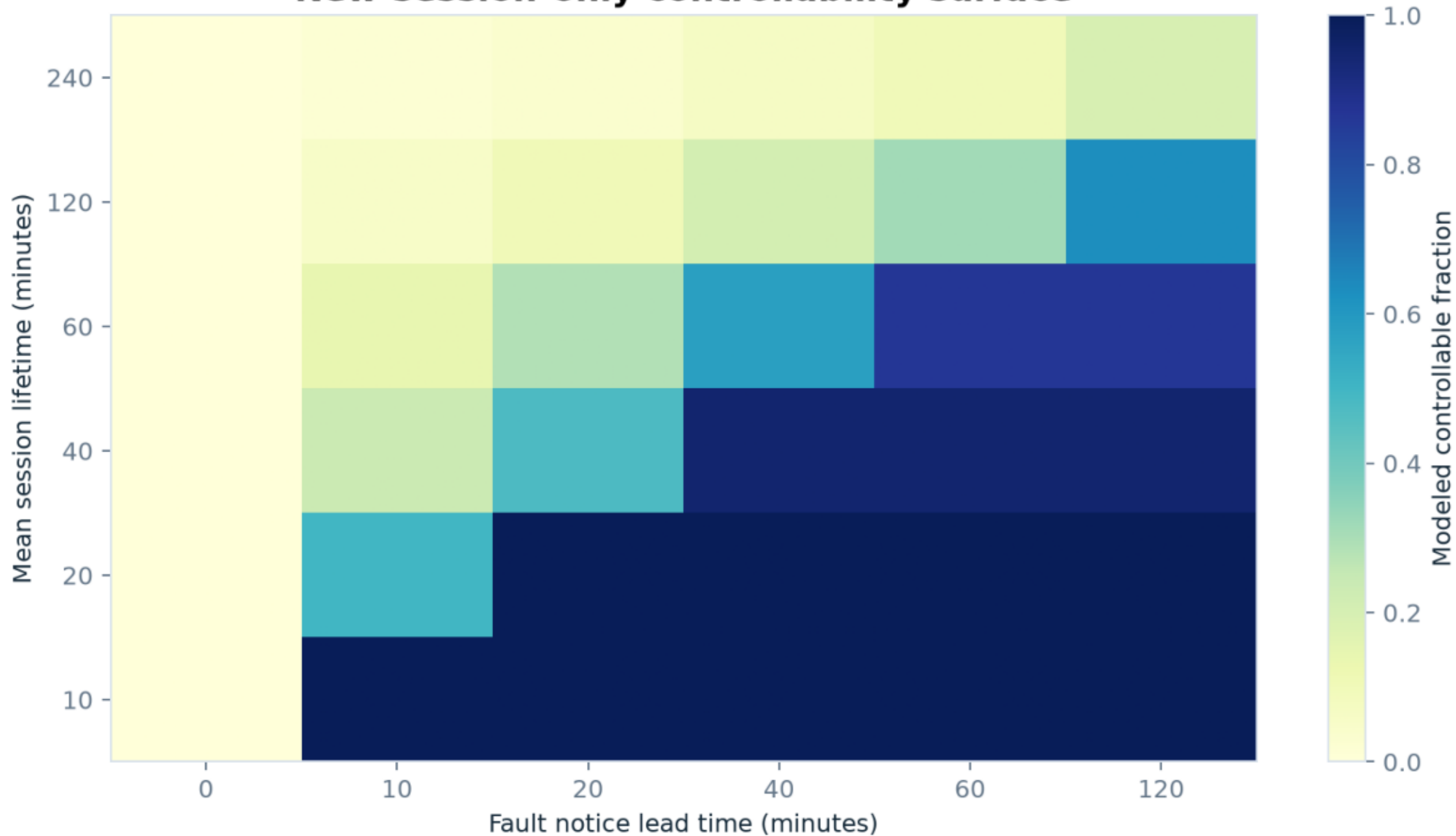


Development seeds 46101–46112 · validation/release untouched

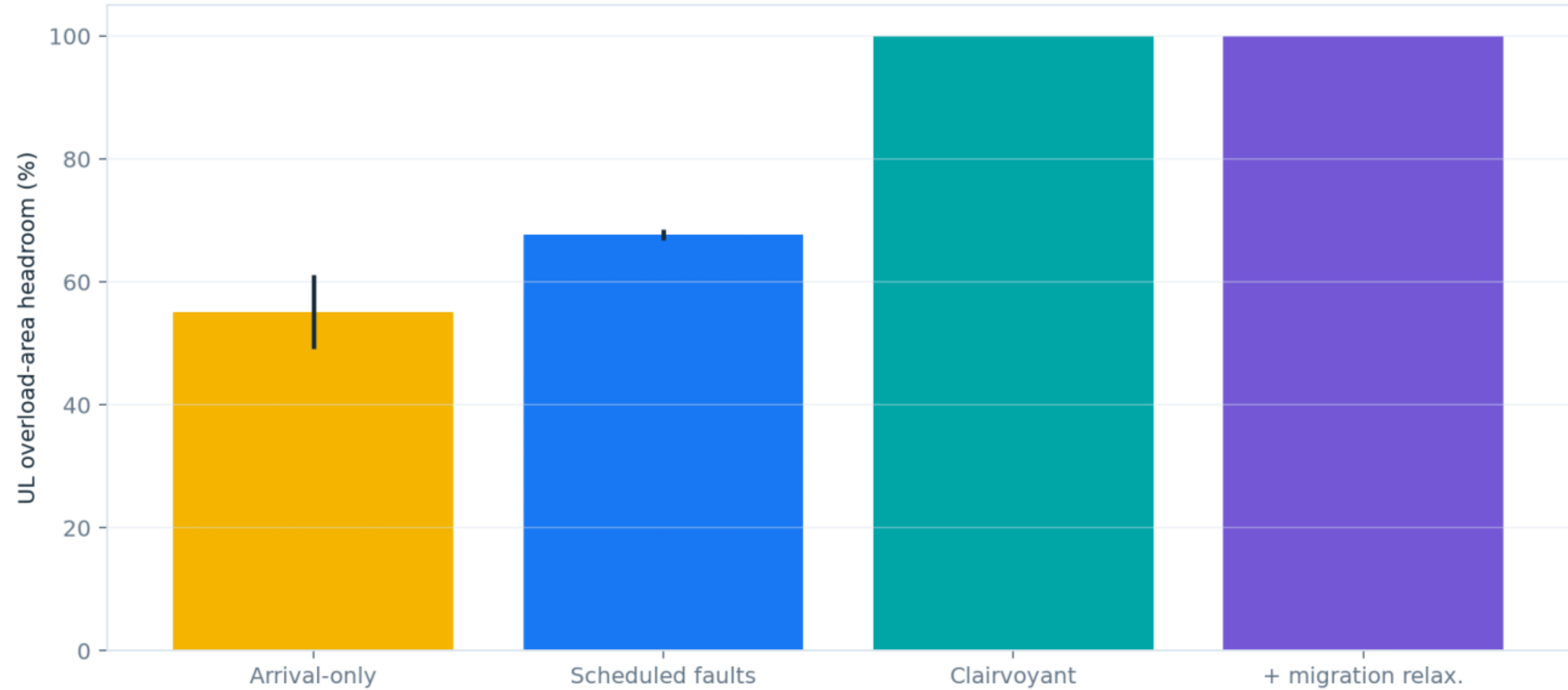
Better forecasts did not automatically improve control



New-session-only controllability surface

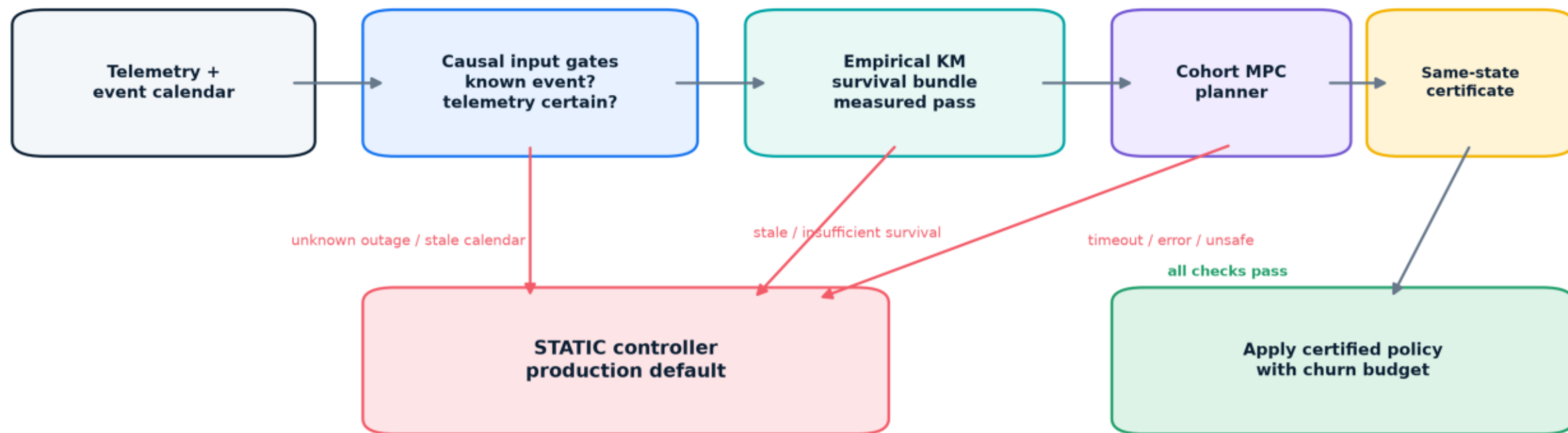


Information ladder: faults dominate controllability



Production architecture: MPC is an optional, guarded branch

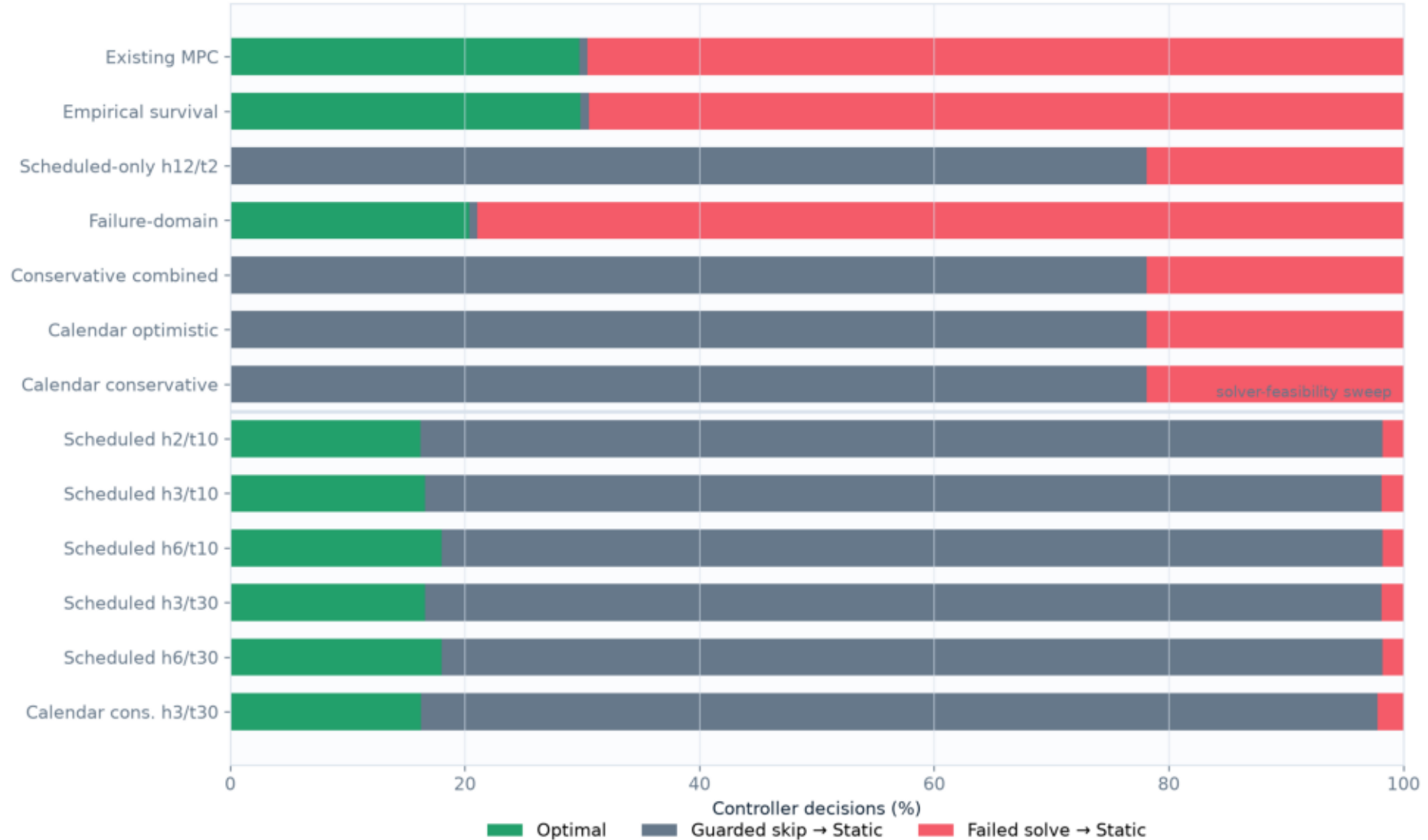
Unknown outages, uncertain telemetry, stale survival, or failed certification return immediately to Static



Measured result: the guardrail plumbing works; no tested MPC branch earned authority to replace Static.

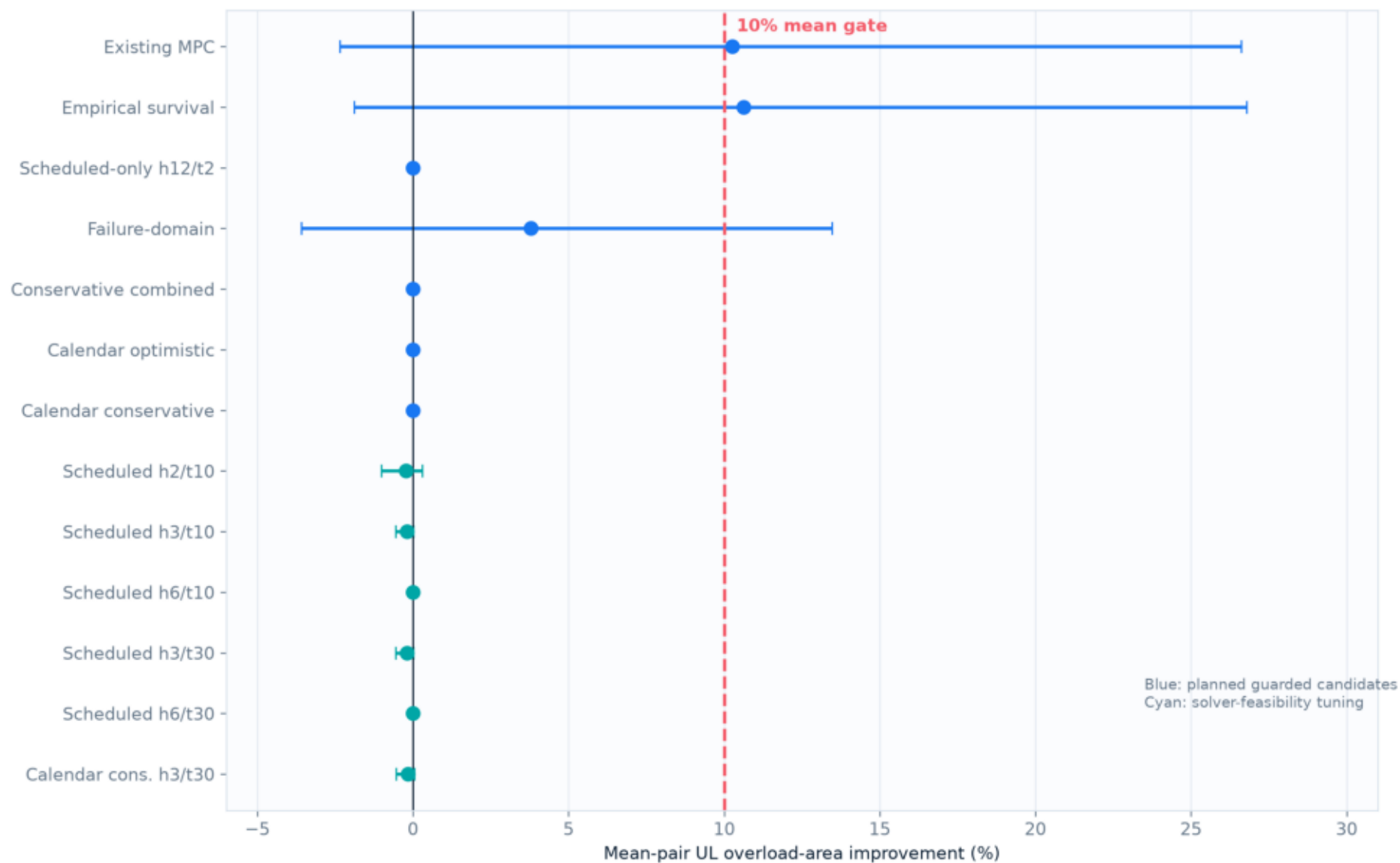
Solver-feasibility tuning removed timeouts—not the performance gap

Share of controller decisions by solver status; failed = timeout, error, or infeasible



All 13 guarded MPC candidates fail the frozen Day-5 bar

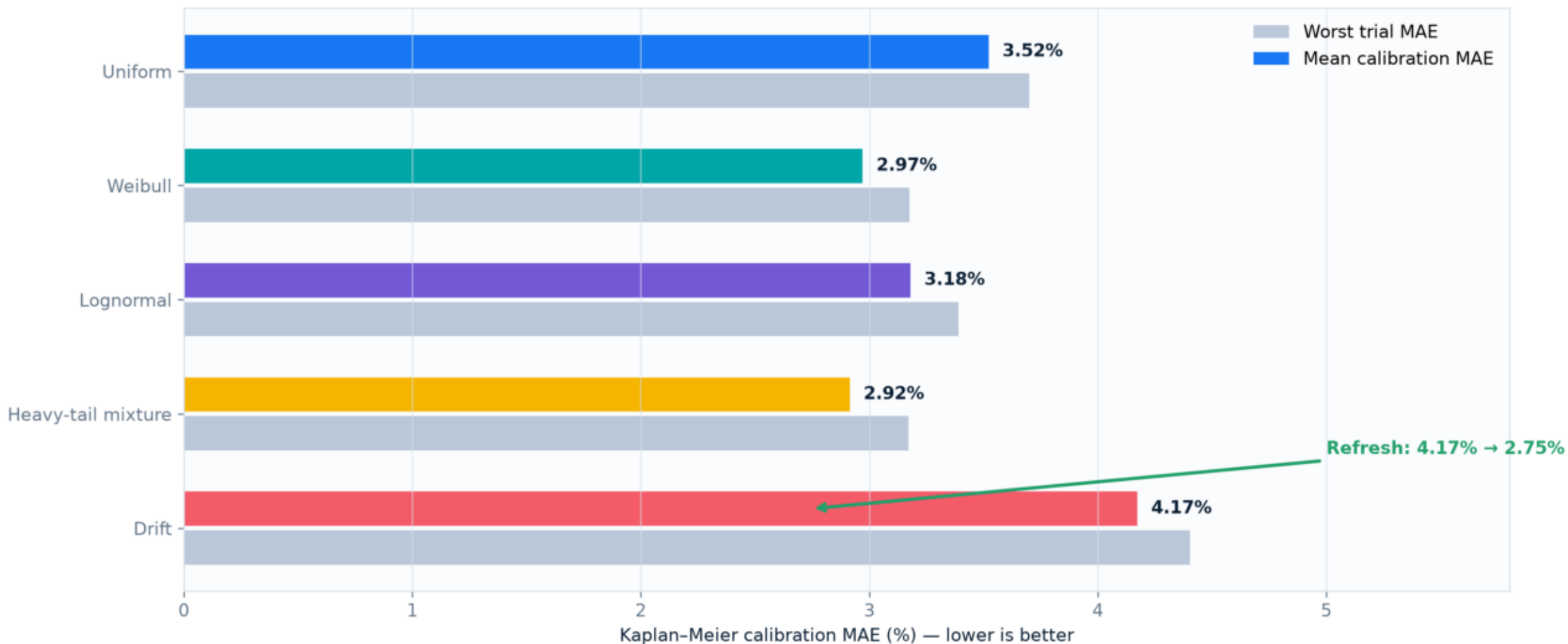
Forest plot shows paired mean UL improvement and bootstrap 95% interval



Development seeds 46101-46112 · validation/release untouched

Distribution-blind lifecycle survival works mechanically

125 auditor-hidden trials · fit sees only start, end, censor, class and timestamps



WORKED

Staleness fail-closed 100% · pooling coverage 25%

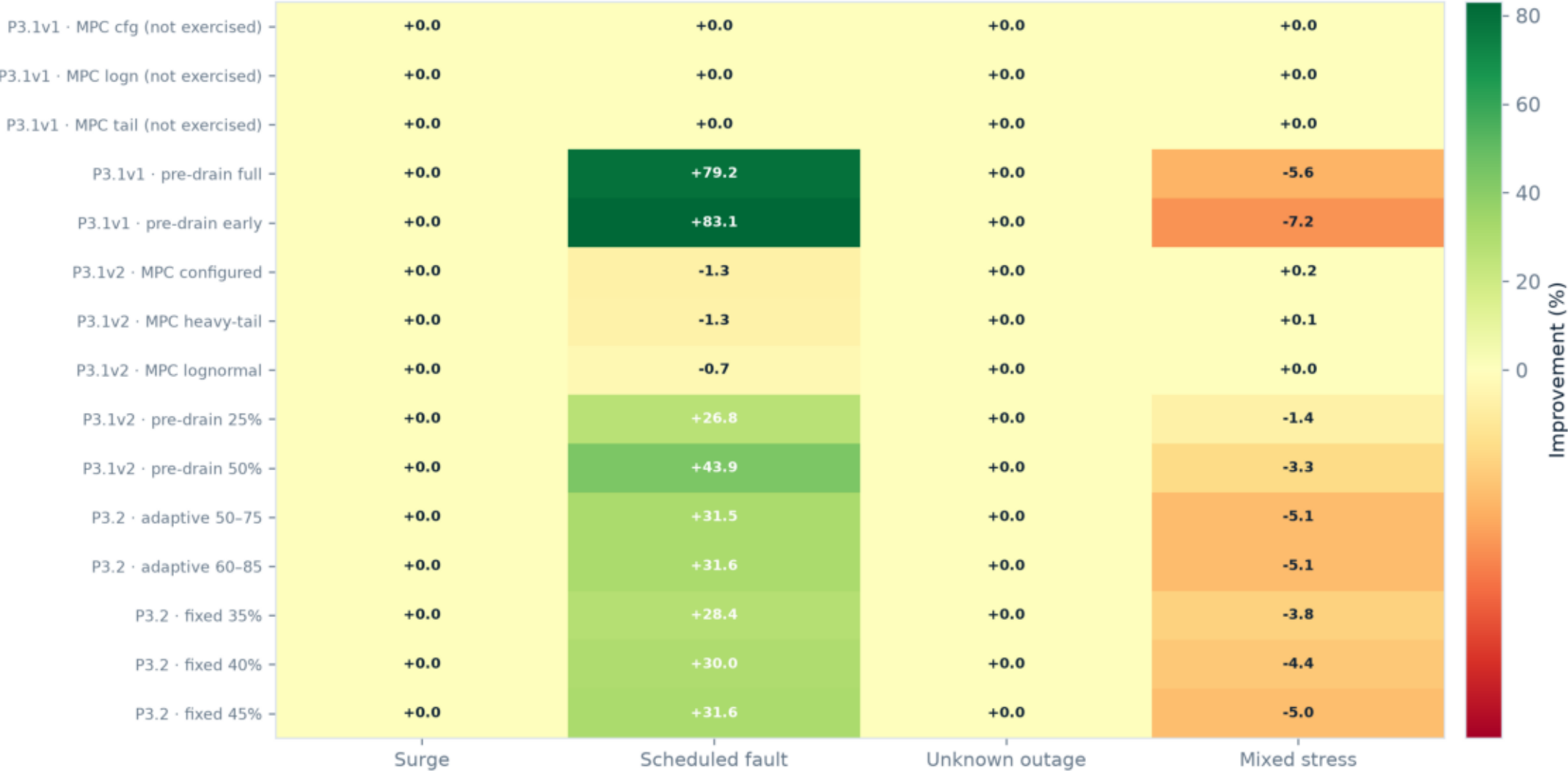
BOUNDARY

Simulated observations $\neq$ external real-world calibration

Fresh development seeds 46401–46472 · validation/release untouched

Where gains came from—and where they failed

Scenario-stratified mean UL-overload improvement (%) · scheduled benefit did not generalize



WORKED: scheduled-fault pre-drain · DID NOT: robust transfer to mixed stress

Fresh development seeds 46401-46472 · validation/release untouched

Pre-drain benefit-tail frontier did not produce a winner

Point size = routing churn · promotion requires mean $\geq 10\%$ and worst pair $> -10\%$

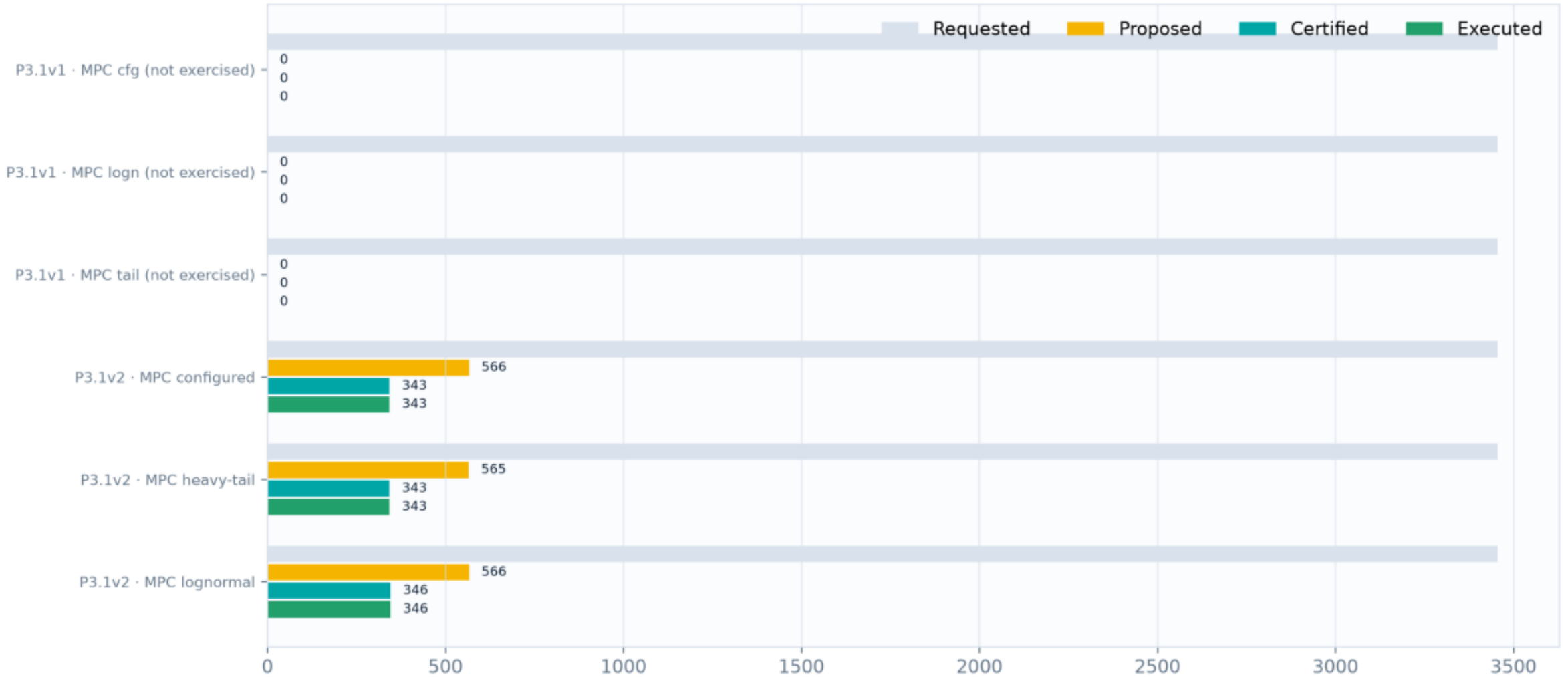


v2 50%: enough mean, tail miss · v2 25% / P3.2: safer, insufficient/uncertain mean

Fresh development seeds 46401-46472 · validation/release untouched

MPC mechanism funnel: v1 was inert; v2 exercised and still did not help

Requested → proposed → certified → executed actions across 24 paired runs per candidate



Decision/action count

WORKED

DID NOT

Preflight enabled 1,697 proposals / 1,032 executions

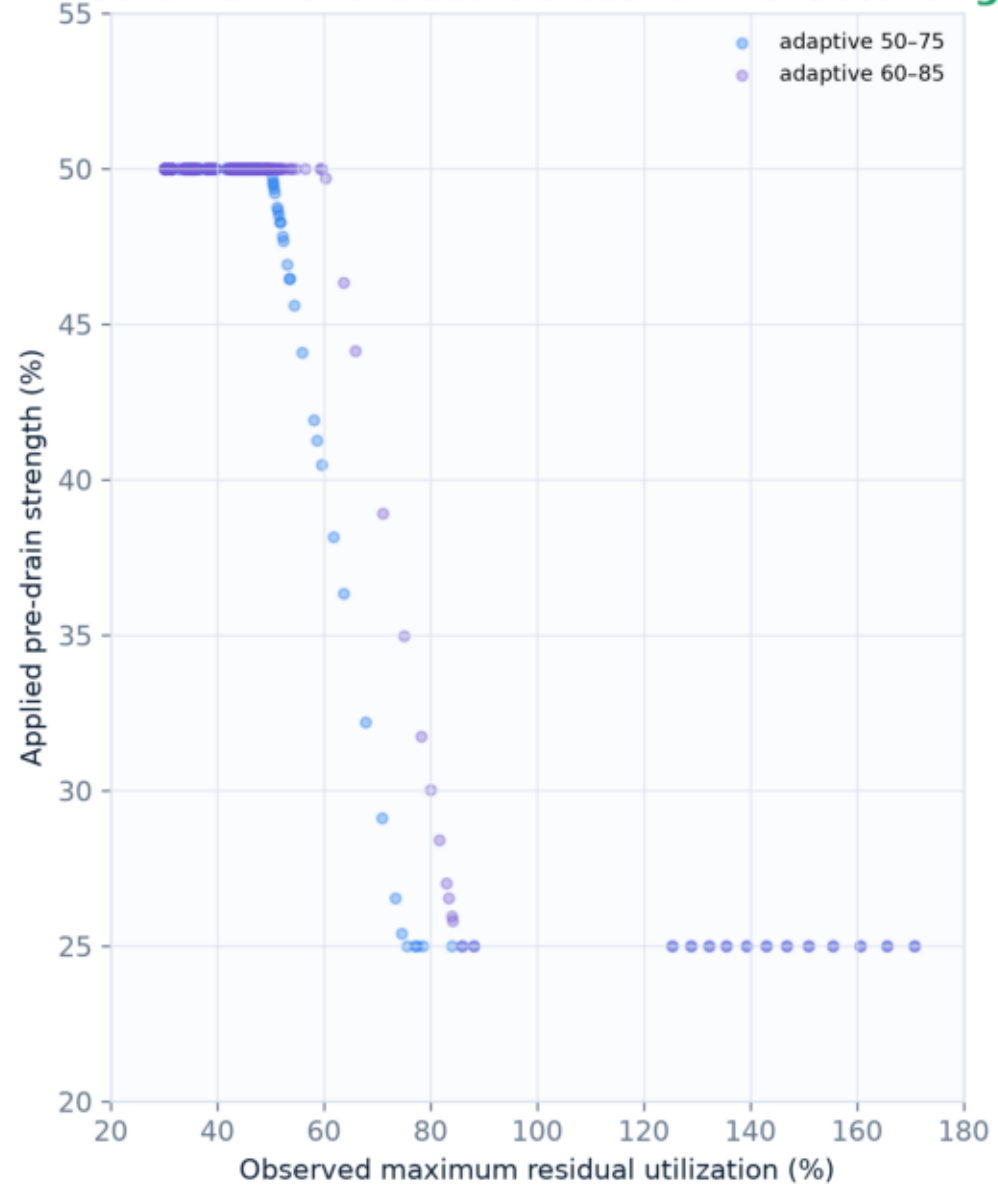
Exercised MPC averaged -0.16% to -0.29%

Fresh development seeds 46401–46472 · validation/release untouched

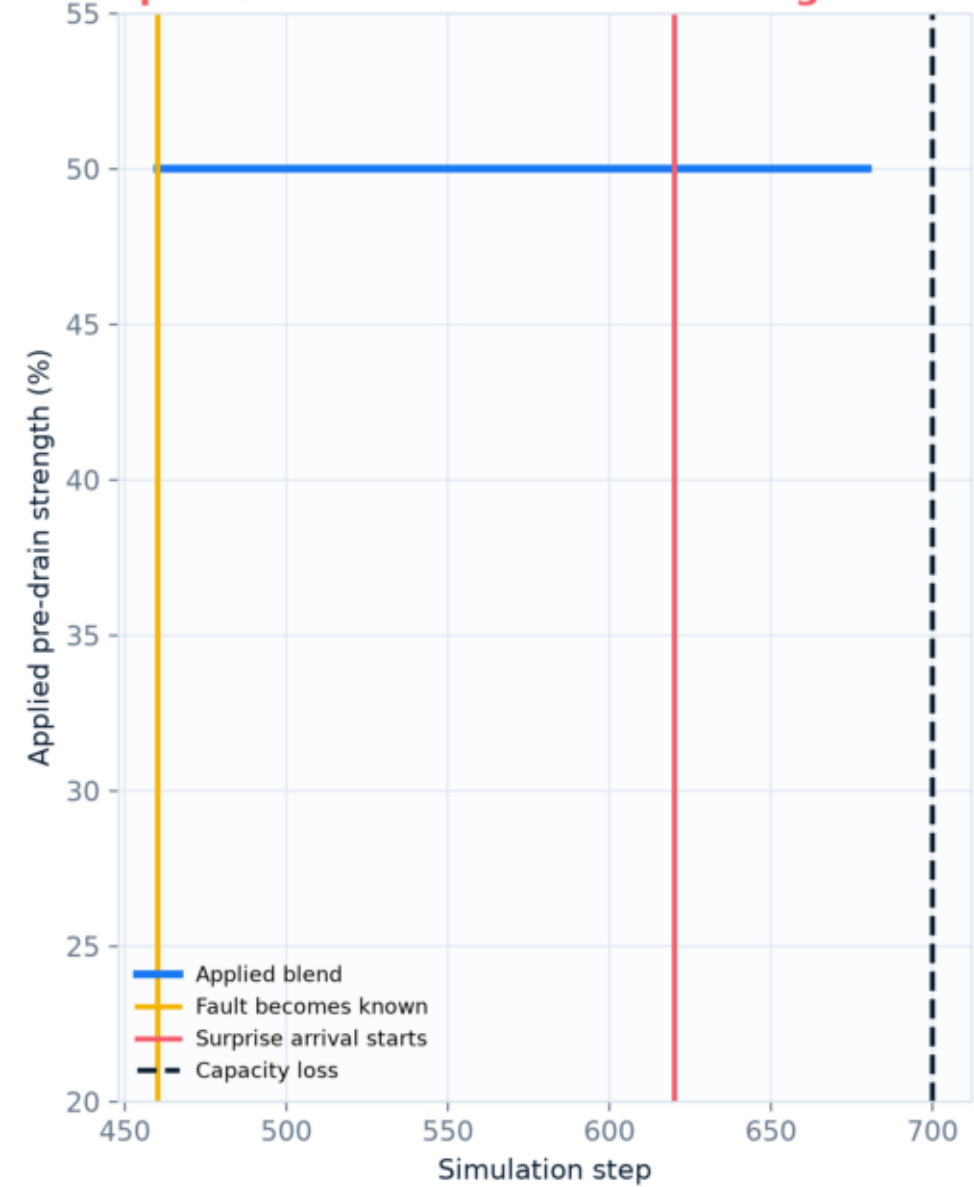
Adaptive pre-drain worked as coded—but causal lag defeated the tail hypothesis

Applied blend responds to observed residual utilization; worst pair's surprise begins after commitment starts

Mechanism exercised across full 25-50% range



Worst pair seed 46468: full 50% throughout lead-up

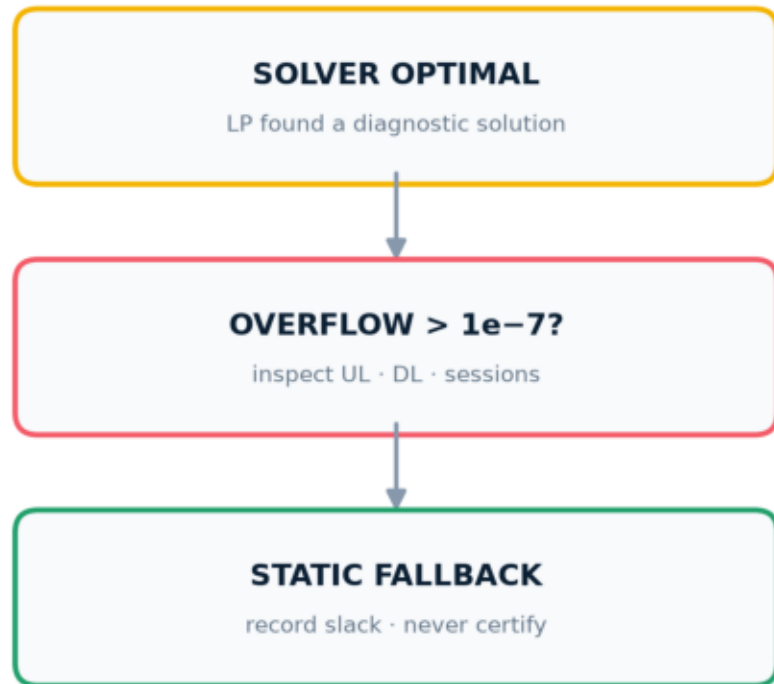
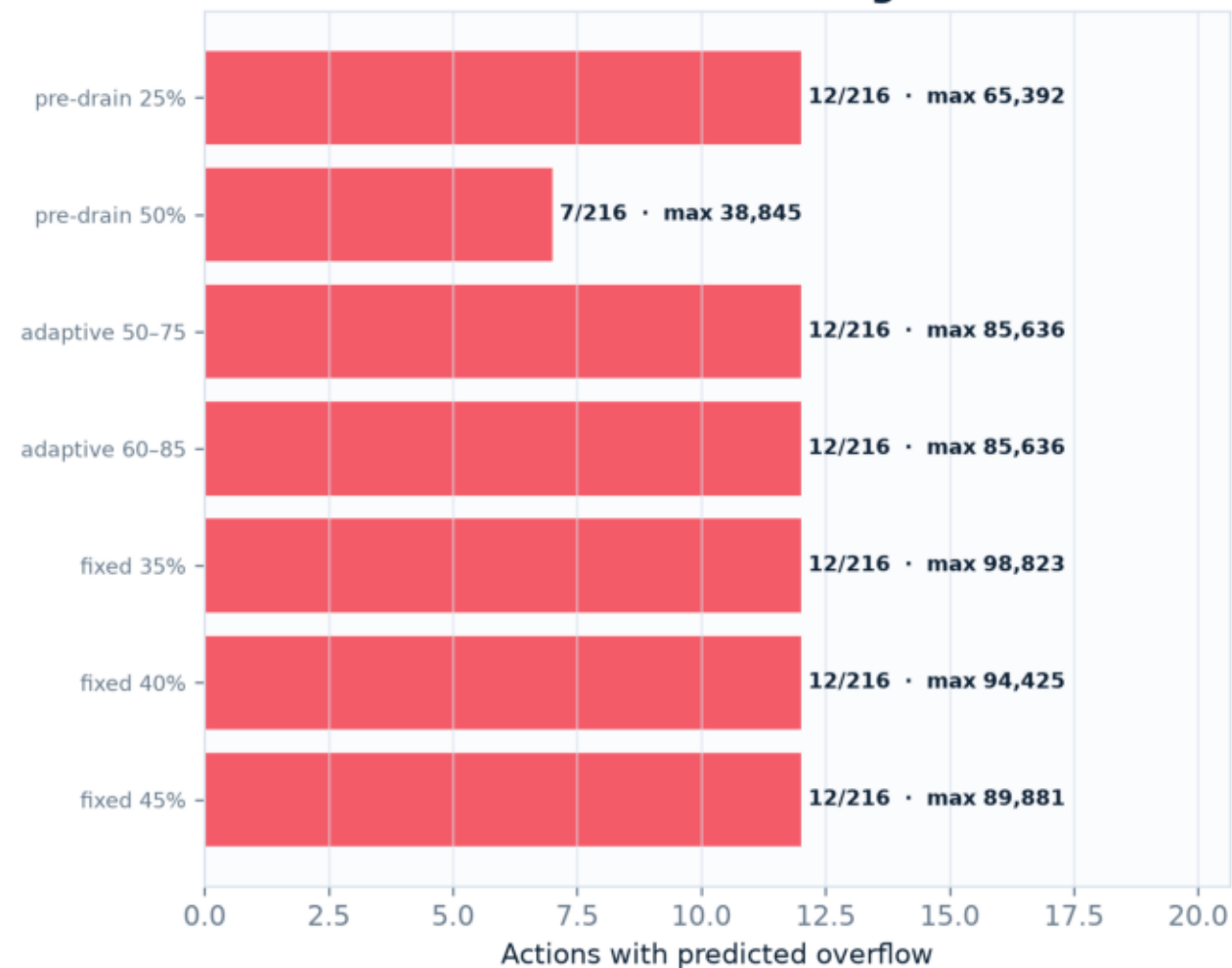


Fresh development seeds 46401-46472 · validation/release untouched

Overflow slack is diagnostic—not a certificate

Historical outcomes stay negative; v6 corrects the controller boundary and the language used to describe it

Recorded historical diagnostics

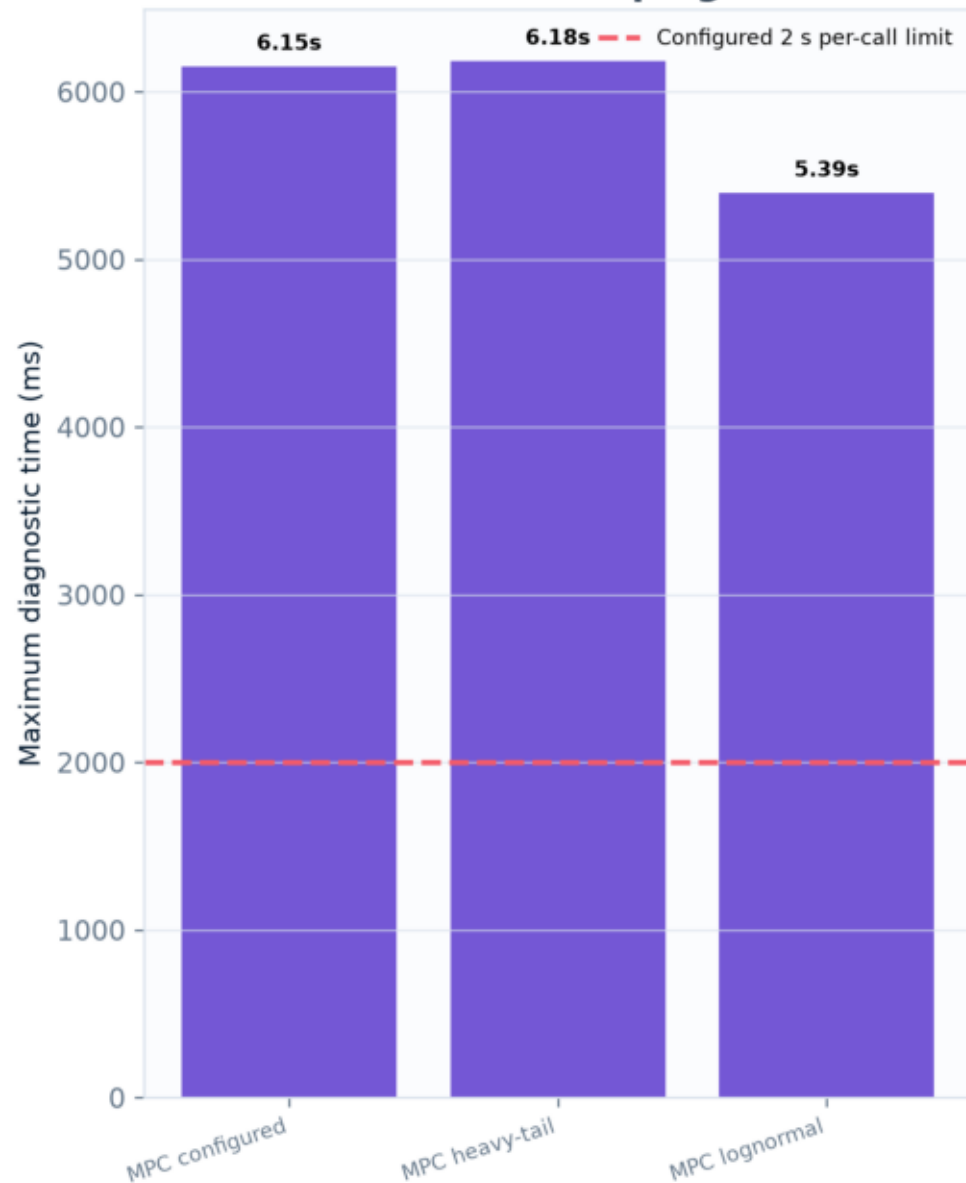


zero_predicted_overflow is now conjunctive

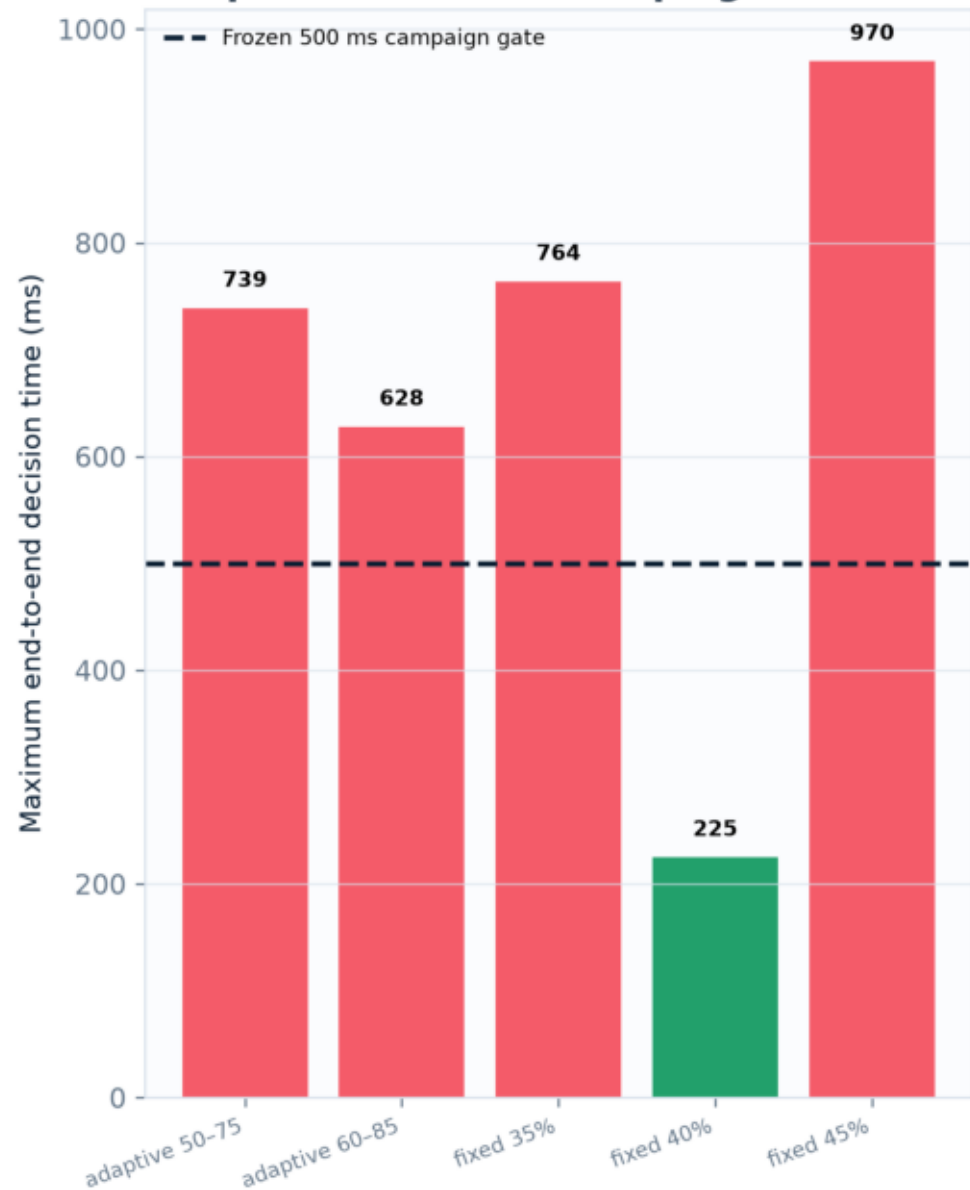
Campaign-saturation latency—not an isolated production benchmark

120 concurrent simulations on a saturated 125-CPU node; useful stress evidence only

P3.1 v2 MPC under campaign saturation



P3.2 pre-drain under campaign saturation

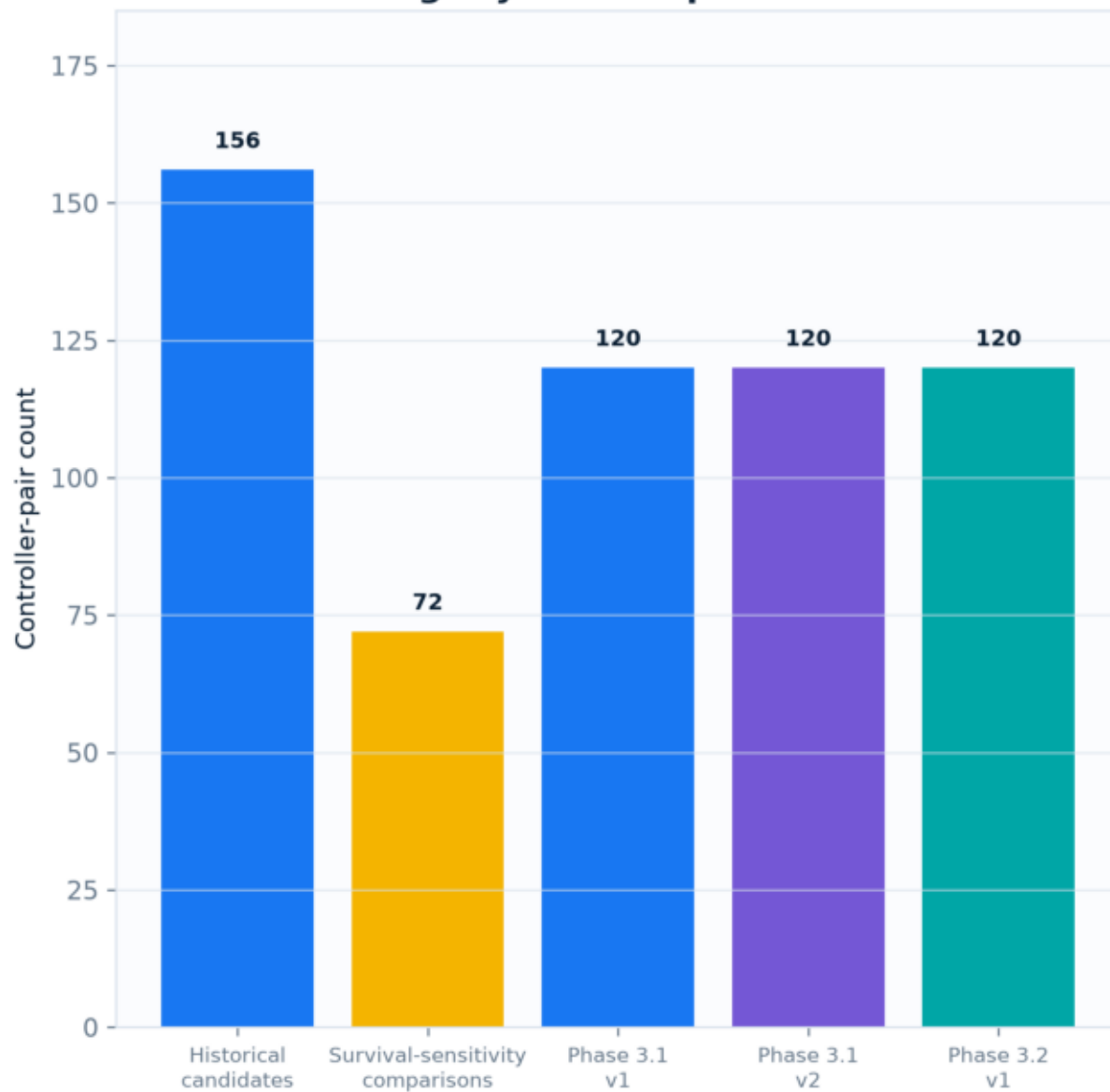


Fresh development seeds 46401-46472 · validation/release untouched

Exact controller-pair inventory and seed firewall

516 declared-candidate pairs + 72 survival-sensitivity comparisons = 588 total

No ambiguity: all 588 pairs classified



Seed firewall held

Development

consumed as registered

46101-46112 · 46401-46472

Validation

UNTOUCHED

46201-46216

Release

UNTOUCHED

46301-46330

Forecast test generated, never evaluated

46003

Fresh development seeds 46401-46472 · validation/release untouched

What worked—and what did not

Post-audit corrections change labels, not outcomes

WORKED



Reproducibility

Historical freezes retained; v6 corrections separately hashed and archived



Lifecycle estimator

Observable export + Kaplan-Meier calibrated across 5 hidden distributions



Overflow safety

Nonzero predicted slack now records by resource and fails closed to Static



Oracle fixture

Authoritative completed-job evaluation restored; full suite is green



Scheduled headroom

Pre-drain repeatedly reduced scheduled-fault overload in replay



Release discipline

516 candidate pairs + 72 sensitivity pairs; zero unsafe promotions

DID NOT WORK



Forecast promotion

Useful gains remained below the frozen 15% target



MPC benefit

Exercised variants were neutral/slightly harmful



Generalization

Scheduled benefit did not transfer safely to mixed stress



Tail safety

Strong blends missed the –10% worst-pair gate



Historical certification

Optimal-with-overflow actions are reclassified as non-certified



Latency claim

225–970 ms reflects saturated campaign execution, not production isolation

FINAL DECISION · RETAIN STATIC · SHADOW/REPLAY ONLY FOR MPC/PRE-DRAIN

Fresh development seeds 46401–46472 · validation/release untouched

What we can defend

- BUILD** A causal, auditable cohort twin that scales to 12 nodes / 384 shards
- MEASURE** Separate forecast, survival, solver, tail, latency, and resource claims
- LEARN** Better prediction and theoretical headroom do not guarantee better control
- DEPLOY** Static remains production; MPC and pre-drain remain shadow/replay

The strongest result is not a promoted algorithm. It is an evidence system that knew when not to promote one.